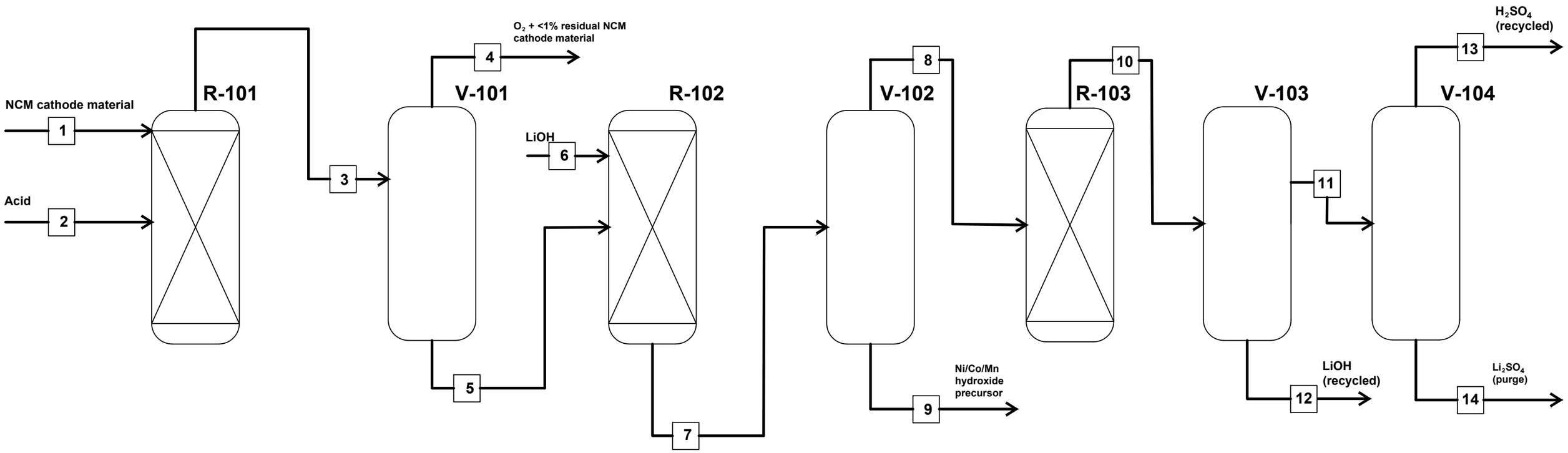

Figure 1: Process Flow Diagram for the Hydrometallurgical Recycling of NCM Cathode Material

Table 1: Stream Table for the NCM Cathode Hydrometallurgical Recycling Process (Figure 1)

Stream Number	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Temperature(°C)	80	80	80	80	80	80	80	80	80	80	80	80	80	80
Pressure(bar)	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Molar Vapor Fraction	0.0	0.0	0.07	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Mass Flow (kg/hr)	100.0	185.0	285.0	9.15	275.9	60.00	335.9	220.9	114.9	220.9	149.5	71.48	119.3	30.20
Molar Flow (kmol/hr)	1.03	3.45	5.28	0.28	4.99	2.51	6.47	5.00	1.47	5.00	2.01	2.98	1.74	0.27
Component Mass Flow Rate (kg/hr)														
NCM Cathode Material	100.0	0.0	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
H2SO4	0.0	142.3	7.6	0.0	7.6	0.0	7.6	0.0	7.6	107.8	107.8	0.0	107.8	0.0
H2O2	0.0	14.2	12.5	0.0	12.5	0.0	12.5	0.0	12.5	0.0	0.0	0.0	0.0	0.0
H2O	0.0	28.5	51.1	0.0	51.1	0.0	51.1	51.1	0.0	11.5	11.5	0.0	11.5	0.0
LiOH	0.0	0.0	8.1	0.0	8.1	60.0	18.9	18.9	0.0	71.5	0.0	71.5	0.0	0.0
Li2SO4	0.0	0.0	37.9	0.0	37.9	0.0	151.0	151.0	0.0	30.2	30.2	0.0	0.0	30.2
Ni(OH)2	0.0	0.0	0.0	0.0	0.0	0.0	66.8	0.0	66.8	0.0	0.0	0.0	0.0	0.0
Co(OH)2	0.0	0.0	0.0	0.0	0.0	0.0	14.3	0.0	14.3	0.0	0.0	0.0	0.0	0.0
Mn(OH)2	0.0	0.0	0.0	0.0	0.0	0.0	13.7	0.0	13.7	0.0	0.0	0.0	0.0	0.0
O2	0.0	0.0	9.1	9.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
NiSO4	0.0	0.0	111.5	0.0	111.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
MnSO4	0.0	0.0	23.3	0.0	23.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
COSO4	0.0	0.0	23.9	0.0	23.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0